Week 02

Concept Learning and Find-S

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Topics to be covered

- Concept learning and target concept
- Instances as attribute-value pairs
- Hypothesis space and candidate rules
- General-to-specific hypothesis ordering
- Least and most specific hypotheses
- Find-S algorithm intuition and steps

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Concept Learning

- Explains how ML learns rules from instances.
- Model moves from training samples → prediction.
- Example: I need loan: Income (**Low**), Credit (**Good**), Job (**Stable**), Defaulter (**No**)
  - Learn the Concept "Should we approve this loan application?"
- Instance is single application having.
- Attribute-value vector: $x = \langle$High, Good, Stable, No$\rangle$
- Label: Correct Decision: Approves (Yes), Rejects (No)
- For concept learning: Label is binary: Yes (1), No (0)
- Benefits:
  - Foundation for supervised classification.
  - Introduces the idea of generalization.
  - Defines the learning goal.

Attribute	Possible values
Income	High, Medium, Low
Credit	Good, Poor
Job	Stable, Unstable
Defaulter	No, Yes

Target Concept

- Instance space (X): Set of all possible instances.
  - $|X| = 3 \times 2 \times 2 \times 2$
  - $|X| = 24$ (possible loan-application)
- Target Concept: Real, unknown positive rule.
  - $c : X \rightarrow \{0, 1\}$
- For each application x, target concept returns:
  - $c\ (x) = 1$: Approved
  - $c\ (x) = 0$: Rejected
  - Rule: Approve only if Credit is Good, Job is Stable, Defaulter is No.

Attribute	Possible values	Possible Instances
Income	High, Medium, Low	3
Credit	Good, Poor	2
Job	Stable, Unstable	2
Defaulter	No, Yes	2

$$c(x) = \begin{cases} 1, & \text{if Credit} = \text{Good, Job} = \text{Stable, and Default} = \text{No} \\ 0, & \text{otherwise} \end{cases}$$

- Learner does not know this rule.

Hypothesis and it's space

- Training example is written as:

$$\langle x, c(x) \rangle$$

- First record is?

$$\langle \langle \text{High, Good, Stable, No} \rangle, 1 \rangle$$

- Hypothesis?
  - One possible rule by model:

$$h : X \rightarrow \{0, 1\}$$

ID	Income	Credit	Job	Default	Label
1	High	Good	Stable	No	Yes
2	Medium	Good	Stable	No	Yes
3	Low	Good	Stable	No	Yes
4	High	Poor	Stable	No	No
5	High	Good	Unstable	No	No
6	Medium	Good	Stable	Yes	No

- Learner does not search every possible rule.
- Searches predefined collection of rules: **hypothesis space**: $H = \{h_1, h_2, \ldots, h_n\}$
- Each hypothesis represented as: $h = \langle a_1, a_2, a_3, a_4 \rangle$
- Each position (a) corresponds to: ⟨Income, Credit, Job, Default⟩
- Symbols used: **?** means any value is accepted, **∅** means no value is accepted.
- Rule: Approve only if Credit is Good, Job is Stable, Defaulter is No.

$$\text{h} = \langle ?, \text{Good, Stable, No} \rangle$$

- ⟨?, Good, ?, ?⟩ means what?

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General vs Specific Rules

- Hypothesis A is general than B if A accepts more rules than B:

$$\mathbf{h_j \geq_g h_k} \Leftrightarrow \forall x \in X, h_k(x) = 1 \Rightarrow h_j(x) = 1$$

- Compare these rules:
  - $h_1 = \langle$High, Good, Stable, No$\rangle$
  - $h_2 = \langle$High, Good, ?, No$\rangle$
- **Why $h_2 \geq_g h_1$?**
- h_1 accepts more rules than h_2, hence $h_2 \geq_g h_1$
- **What is the most general hypothesis?**
  - $h_b = \langle ?, ?, ?, ? \rangle \Leftrightarrow \forall x \in X, h_b(x) = 1$
- **What is the most specific hypothesis?**
  - $h_a = \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle \Leftrightarrow \forall x \in X, h_a(x) = 0$
- **Goal:** Find out the useful learned rules, lie between the two extremes.
- **Challenge: Which one is more general?**
  - $h_A = \langle ?$, Good, Stable, No$\rangle$, $h_B = \langle$High, ?, Stable, ?$\rangle$
  - $h_A \not\geq_g h_B$ and $h_B \not\geq_g h_A$ because of the partial order, not straight order.

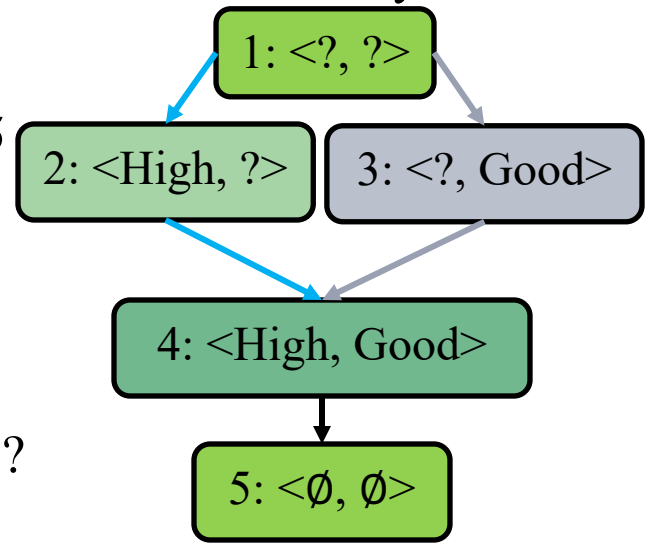
ID	Income	Credit	Job	Default	Label
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6	Medium	Good	Stable	Yes	No

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General → Specific Path

- Single straight line: Most general → General → Specific → Most Specific
- Often not possible: $h_A = \langle ?, \text{Good}, \text{Stable}, \text{No} \rangle$, $h_B = \langle \text{High}, ?, \text{Stable}, ? \rangle$
- So, hypothesis is not always more general than the other.
- This is called partial ordering.
- Consider small example: Income and Credit only:
- Possible paths:
- Straight line: 1 → 2 → 4 → 5
- Not possible path?
  - 1 → 2 → 3 → 4 → 5
- Can we use BFS?
- What are the issues with BFS?
- Time complexity of BFS?
  - $O(V + E) = O(109 + 321) = O(430)$

ID	Income	Credit	Label
1	High	Good	Yes
2	Medium	Good	Yes
3	Low	Good	Yes
4	High	Poor	No
5	Medium	Poor	No
6	Low	Poor	No



Think?

- BFS explores all the possible rules; hence, the time complexity is huge.
- Which rule should be selected?
  - The most general? $h_b = \langle ?, ?, ?, ? \rangle$
  - The most specific? $h_a = \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$
  - In-between rule? Which attributes are important?
- How do you find the in-between rule?
- What if one focuses only on negative rules?
- What if one focuses only on positive rules?
  - **Yes!** Positive rules are required only in this situation.
  - For a hypothesis h and instance $x \in X$:

$$h(x) = \begin{cases} 1, & \text{if } x \text{ satisfies every constraint in } h \\ 0, & \text{otherwise} \end{cases}$$

  - Pseudocode?

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Find-S Algorithm

- Start with the most specific hypothesis.
- For each positive training example:
  - Compare it with the current hypothesis.
  - Keep matching attribute values unchanged.
  - Replace each mismatching value with ?.
- Return the final hypothesis.
- Goal: Find-S expands its rule only when a positive example forces it to do so.
- Strengths:
  - Easy to understand.
  - Shows how a learner can generalize from examples.
  - Produces a simple, maximally specific hypothesis.
- Limitations
  - Ignores negative examples.
  - Assumes data has no mistakes or noise.
  - Returns only one rule, even if many rules fit the data.

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Example

- Start: most specific rule
  - $h_0 = \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$
- After first positive example
  - $h_1 = \langle$High, Good, Stable, No$\rangle$
- After second positive example
  - $h_2 = \langle$?, Good, Stable, No$\rangle$
- After third positive example
  - $h_3 = \langle$?, ?, Stable, No$\rangle$
- Income and Credit differs, so generalize both.
- Approve all loans When:
  - Employment is Stable, and Default status is No.
- Time complexity of Find-S Algorithm?

  For each of ***n*** training examples:

  Check or update each of ***m*** attributes
  - $n \times m = O(nm) = ?$

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Recap

- **Instance:** one example represented by attribute-value pairs.
- **Target concept:** the unknown true classification rule.
- **Hypothesis:** one candidate rule.
- **Hypothesis space:** all candidate rules allowed by the representation.
- **General hypothesis:** accepts more examples.
- **Specific hypothesis:** accepts fewer examples.
- **Find-S:** starts very specific and generalizes to cover positive examples.

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Thank You!